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Review

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Advancements in humanoid robot dynamics and learning-based locomotion control methods

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Abstract

Humanoid robots are attracting increasing global attention owing to their potential applications and advances in embodied intelligence. Enhancing their practical usability remains a major challenge that requires robust frameworks that can reliably execute tasks. This review systematically categorizes and summarizes existing methods for motion control and planning in humanoid robots, dividing the control approaches into traditional dynamics-based and modern learning-based methods. It also examines the navigation and obstacle-avoidance capabilities of humanoid robots. By providing a detailed comparison of the advantages and limitations of various control methods, this review offers a comprehensive understanding of current technological progress, real-world application challenges, and future development directions in humanoid robotics. Key topics include the principles and applications of simplified dynamic models, widely used control algorithms, reinforcement learning, imitation learning, and the integration of large language models. This review highlights the importance of both traditional and innovative approaches in advancing the adaptability, efficiency, and overall performance of humanoid robots.

Keywords: Humanoid robot, locomotion control, dynamics and machine learning, path planning

1. INTRODUCTION

Amid rapid technological advancements, humanoid robots have emerged as a key frontier in artificial intelligence and robotics, attracting significant research interest in recent years. Humanoid robots, characterized by their human-like form and functionality, have vast potential in industrial automation, healthcare, home

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services, and education. They also play irreplaceable roles in extreme or hazardous environments, such as planetary exploration, firefighting, and nuclear radiation scenarios[1–5]. These diverse applications not only enhance the quality of life and improve work efficiency but also address unique challenges, highlighting their broad significance[6].

Inspired by the human body, humanoid robots typically consist of a head, torso, two arms, and two legs, which allow them to perform human-like actions and tasks. Their development is a highly interdisciplinary endeavor that integrates mechanical design, sensor technology, motion control, and artificial intelligence. These technologies collectively enable humanoid robots to operate safely and effectively across various scenarios, including industrial production, medical procedures, rehabilitation, disaster response, and search-and-rescue operations.

Motion control is a fundamental aspect of humanoid robot technology that directly affects movement efficiency and operational safety. Effective control strategies enable robots to follow predefined gaits precisely and respond in real time to sudden changes or disturbances in balance. Key technical challenges include accurately modeling the dynamic characteristics of human walking, coping with environmental uncertainties, and optimizing the control algorithms. Addressing these challenges will improve both the functionality and safety of humanoid robots in complex environments.

Currently, electric motor drives dominate the control of humanoid robots. Hydraulic-driven robots, such as Boston Dynamics' Atlas[7], which were once praised for their performance, have been retired, and research focus has shifted to electric actuation. In recent years, the field has witnessed rapid progress driven by emerging research initiatives and technological innovations.

The development of electric humanoid robots began with Japan's ASIMO[8], which features 34 degrees of freedom, stands 130 cm tall, weighs 54 kg, and performs smooth walking, running, jumping, and stair climbing. In 2022, Tesla unveiled its Optimus prototype[9], which was capable of single-leg balance and was equipped with self-correcting limbs. Agility Robotics' Digit, designed for warehouse logistics, is scheduled for commercial deployment in 2024. Figure AI launched Figure 02, a general-purpose humanoid robot integrated with ChatGPT for language understanding and task execution. China has also made significant advances, for example, Unitree's H1, which is widely deployed in universities, and Fourier Intelligence's GR-1, which features a unique structure optimized for bipedal locomotion and manipulation.

Although prior reviews have addressed reinforcement learning (RL) and teleoperation in humanoid robots, a comprehensive review specifically focusing on motion control and path planning is still lacking. This study aims to fill this gap by summarizing recent progress, highlighting motion control frameworks, and exploring future trends. The main contributions of this study are as follows: (1) A systematic review of dynamics-based and learning-based control methods for humanoid robots; (2) A detailed comparison of these approaches, emphasizing their strengths, limitations, and features; (3) A summary of path planning and obstacle avoidance techniques applicable to humanoid robots; (4) An analysis of existing challenges and prospects for future research.

The remainder of this paper is structured as follows: Section 2 introduces dynamics-based control methods; Section 3 reviews learning-based control approaches; Section 4 summarizes path planning and obstacle avoidance techniques; and Section 5 discusses current limitations and future directions. Finally, Section 6 concludes the paper.

2. DYNAMICS-BASED CONTROL METHODS

Control methods based on dynamic models utilize the robot motion equations and dynamic constraints to design precise and reliable control strategies. By formulating the control problem within this dynamic frame-

Table 1. Common simplified models of humanoid robots

Model	Works	Characteristic
Cart-table model	[10], [11]	When the trolley moves at a specific acceleration, the table can maintain instantaneous balance and fall
LIP	[12], [13], [14], [15], [16]	The 2D LIP can only generate the walking trajectory in the 2D plane; the 3D LIP is generalized to the 3D space based on the 2D LIP, and their center-of-mass height is constant
LIPPF	[17], [18], [19], [20], [21]	LIPPF simulates the angular momentum of an object's CoM using a flywheel with a center-of-mass moment of inertia and rotation angle restrictions instead of a single particle
SLIP	[22], [23], [24], [25]	SLIP models can better simulate the vibration of the CoM and the smooth motion of the legs to adapt to the control of a humanoid robot in a high-speed running mode, such as jumping and trotting

LIP: Linear inverted pendulum; 2D: two-dimensional; 3D: three-dimensional; LIPPF: linear inverted pen-dulum plus flywheel; CoM: center of mass; SLIP: spring-loaded inverted pendulum.

work, optimal control inputs can be computed to ensure that the robot achieves stable and desired motion under various conditions. This chapter introduces several widely used simplified dynamic models for humanoid robots, including the linear inverted pendulum model (LIPM), cart-table model, and spring-loaded inverted pendulum (SLIP) model. It also reviews common control methods based on these models, such as proportional–integral–derivative (PID) control, model predictive control (MPC), and zero moment point (ZMP) control, and discusses their roles in achieving effective and stable humanoid locomotion.

2.1. Simplified dynamic models of bipedal walking

Approximation is a practical approach for managing the complexity of humanoid-robot motion control. Complex robot models are often simplified to enable efficient computation. Several dynamic models have been widely adopted for bipedal gait planning, with the cart-table model and various inverted pendulum models being the most common.

Among these, two-dimensional (2D) and three-dimensional (3D) LIPMs are frequently used because of their balance between simplicity and accuracy. The 3D linear inverted pendulum plus flywheel (LIPPF) model extends this by incorporating the flywheel rotational inertia, offering a more complete depiction of dynamic stability. The SLIP model further introduces elastic components to mimic the spring-like behavior observed in natural locomotion, highlighting energy-efficient gait strategies.

The following sections examine the principles and features of these simplified models, discussing their contributions to humanoid gait planning and practical implementation in locomotion control. A brief overview is provided in Table 1.

2.1.1. Cart-table model

The cart-table model[10] is a simplified representation commonly used to approximate the dynamics of bipedal-walking robots. In this model, the robot is abstracted as a massless table supported by a movable cart on a horizontal plane, which captures the fundamental principles of balance and motion. When the cart accelerates appropriately, the table remains balanced without tipping, which is a key behavior for understanding bipedal stability.

This model reduces the complex dynamics of bipedal walking to a 2D analysis. As shown in Figure 1, the robot operates in the z-x plane, with the x-axis representing the forward motion and the z-axis perpendicular to the ground. Movement of the cart along the x-axis shifts the force distribution on the table, determining the location of the center of pressure (CoP), also known as the ZMP.

The ZMP is the point at which the net moment from the gravitational and inertial forces is equal to zero. By maintaining the ZMP within the support polygon - the area between the robot's feet - the robot achieves stable walking and avoids falling over. This model provides a foundational framework for balance control and guides

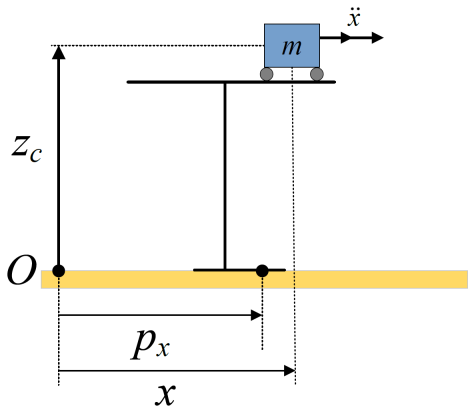


Figure 1. Cart-table model.

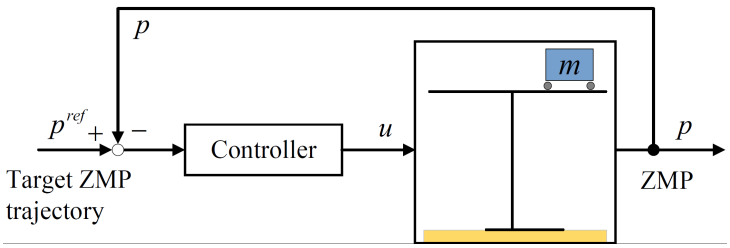


Figure 2. Principle of walking pattern control based on ZMP. ZMP: Zero moment point.

the development of more advanced locomotion algorithms for humanoid robots.

The ZMP based on this model is given as follows:

$$p = x - \frac{z_c}{g}\ddot{x} \tag{1}$$

ZMP has become an essential criterion for analyzing the dynamic balance. Figure 2 illustrates the principle of walking pattern generation based on ZMP. Given the target ZMP trajectory, the required center of mass (CoM) motion is calculated. Based on the calculated CoM trajectory, the actual ZMP trajectory can be computed. The overall goal is to minimize the error between the actual ZMP trajectory and the given ZMP trajectory to enhance the robot stability during motion (as deviations may lead to falls). This can be achieved through feedback control to track the target of the ZMP trajectory. In this control system, the input is the target position of the ZMP, denoted as p^{ref}. The controller uses the feedback input of tracking errors to output an acceleration rate u for the CoM, which serves as the input to the actuator. The actuator determines the actual ZMP and CoM coordinates based on the system's state transition and motion equations. Using this control system, it is possible to generate the motion trajectory of the CoM while ensuring that the ZMP tracks the predefined reference position[11].

2.1.2. LIPM

The 2D LIPM proposed by Kajita *et al.*[12] confines the robot dynamics to the sagittal (x–z) plane. By disregarding lateral motion, it models only the forward–backward translation and small vertical oscillation of the CoM and remains highly effective for basic gait planning. The 3D LIPM extends this formulation to include the lateral (y) degree of freedom, yielding more natural and stable walking patterns, and thus is preferred for full-body humanoid control. The 2D and 3D LIPM models share common characteristics, which can be summarized as follows: (1) All mass is lumped at the hip (CoM); (2) The legs are massless telescopic links; (3) The

CoM height is constant.

Although the constant-height assumption linearizes the equations and simplifies the control design, it restricts gait versatility. Human biomechanics show that CoM height varies during locomotion, enabling faster movement[26]. To relax this constraint, Zhang *et al.* introduced a wheel–leg switching strategy[27], and Shafii *et al.* developed a variable-height LIPM with a dedicated height-trajectory generator[14]. Simulations confirmed that variable-height gaits achieved higher walking speeds than fixed-height gaits, broadening the applicability of the LIPM framework.

2.1.3. LIPPF

Human motions, such as forward lunges or rapid arm swings, exploit angular momentum to maintain balance. The standard LIPM neglects rotational inertia and therefore cannot capture these behaviors. Pratt *et al.* addressed this shortcoming using the LIPPF, which replaces the point mass with a flywheel that embodies both the CoM inertia and bounded rotational angles[17]. Consequently, the LIPPF reproduces the angular-momentum effects essential for human-like movement.

The key assumptions of the LIPPF are as follows: (1) All mass is lumped at the flywheel center (CoM); (2) The legs are massless telescopic rods; (3) The flywheel is free to rotate; therefore, its inertia can counteract the pendulum-like sway. Increasing this inertia improves stability and better mimics the human gait[18,19]. Several studies have further relaxed the constant height constraint on the CoM[20,21], yielding an even more accurate dynamic model that enables humanoid robots to walk more stably and move similarly to humans.

2.1.4. SLIP

In his study[22] of terrestrial biological jumping movements, Blickhan discovered that various insects and animals - whether jumping, trotting, or running at high speeds - exhibit dynamics closely matching those predicted by the SLIP model. The SLIP model excels at simulating the oscillations of the CoM and compliant behavior of leg movements, making it particularly suitable for controlling humanoid robots in high-speed locomotion modes, such as jumping and running.

1) The characteristics of the SLIP model can be summarized as follows:
2) All of the robot's mass is concentrated at its hip position, simplifying the dynamics to focus on leg behavior. Legs exhibit different behaviors during different phases of movement.
    - During the single support phase, the supporting leg maintains a constant length.
    - During the double support phase, both legs undergo elastic extension and contraction, mimicking the spring-like behavior observed in natural locomotion.
3) The CoM can move vertically, allowing the model to capture vertical oscillations typical of dynamic movements such as jumping and running.

An increasing number of researchers have applied the SLIP model to study dynamic locomotion in legged robots. Wensing and Orin[23] were the first to embed the SLIP model into a full-body humanoid model, simulating humanoid robots running stably at speeds of up to 6.5 m/s, which is comparable to the speeds of Olympic athletes in the 5,000-meter race. Xiong *et al.* utilized a hybrid linear inverted pendulum (H-LIP)-based stepping controller to perturb periodic walking motions based on the 3D-driven spring-loaded inverted pendulum (3D-aSLIP) model[24]. They integrated the dynamics of the 3D-driven SLIP model into the Atlas robot, successfully generating versatile walking behaviors, including periodic walking, fixed-position tracking, trajectory tracking, and walking with varying heights. This method provides a valuable reference for generating dynamic and versatile humanoid walking patterns with nonconstant CoM heights. Additionally, the SLIP model can be used to generate various motion patterns, such as walking, running, and walking-to-running transitions[25].

2.1.5. Flexible linear inverted pendulum model

The flexible linear inverted pendulum model (FLIPM)[15,16,28] enhances the basic LIPM by incorporating additional elements, such as springs and dampers. These components are used to simulate the flexibility of links, joints, and servomotors, which are essential for capturing the more nuanced and realistic dynamic behavior of humanoid robots. By adding these flexible elements, the FLIPM can accurately represent the compliant and adaptive nature of human-like movements.

This model accounts for the elasticity and damping characteristics inherent in real robotic systems, allowing a more detailed analysis of how these factors influence balance, stability, and motion control. The incorporation of flexibility enables the FLIPM to better handle dynamic interactions with the environment, such as walking on uneven surfaces or responding to unexpected perturbations. This leads to improved performance in tasks that require a high degree of adaptability and precision. Overall, FLIPM provides a more comprehensive and realistic framework for studying and designing control strategies for humanoid robots, bridging the gap between simplified theoretical models and the complex realities of physical robotic systems.

2.2. Control methodology

Based on a dynamic model, it is essential to design suitable control systems and algorithms to drive the robot joint actuators, ensuring balance and effective motion execution. Common control algorithms for humanoid robots include PID control[29–31], which is simple and effective for precise joint position and velocity control; MPC[32–35], which predicts future states using a robot dynamics model to optimize control inputs; ZMP control[36–38], which is crucial for maintaining stable walking by keeping the ZMP within the support polygon; and inverse kinematics (IK)[39–41] and inverse dynamics (ID)[42,43], which compute the necessary joint angles and torques for desired motions. Additionally, central pattern generators (CPGs) generate rhythmic outputs for cyclic movements such as walking and running. Together, these algorithms improve the robot's ability to perform complex tasks with stability and precision.

2.2.1. PID control

PID control features a simple structure, strong robustness, and adaptability. In the field of industrial control, PID control algorithms are widely applied[44]. A PID controller is a linear controller that computes an error signal e(t) as the difference between a desired setpoint $r(t)$ and the actual output $y(t)$, $e(t) = r(t) - y(t)$. The controller adjusts the controlled object by combining proportional (P), integral (I), and derivative (D) actions of the error through linear combinations.

From a temporal perspective, the proportional action addresses the current error, the integral action accounts for the error's history, and the derivative action reflects the error's rate of change. Together, these components provide a balanced approach to managing past, present, and future system states[45–47].

Alasiry *et al.* applied a PID control system to the humanoid dance robot ERISA, enabling it to walk stably on sloped surfaces without falling[48]. Analog tilt sensors were installed on the robot's shoulders to measure terrain slope. Using the measured tilt values, they adjusted the robot's servo control equations via PID to automatically correct its posture opposite to the downward slope. The walking trajectories consist of gait position data sets. The study established an inverted pendulum model, from which IK computed joint angle setpoints θ_{set}. As the robot walks on slopes, the shoulder tilt sensor measures the actual body tilt angle θ_{tilt}. The joint angle command θ_{act} is set based on the difference between θ_{set} and θ_{tilt}, causing the servo to rotate and compensate for the tilt. The simulation results demonstrated that PID balance control with tilt sensor feedback effectively stabilized ERISA, allowing steady walking on sloped surfaces.

2.2.2. Hybrid zero dynamics

Hybrid zero dynamics (HZD) has become a popular framework for dynamic and underactuated bipedal walking. The principle of HZD is based on the analysis and control theory of nonlinear dynamical systems. The core idea is to transform the dynamic properties of the robot into a zero-dynamic system and design controllers

to achieve these zero dynamics. For a system with constraints $y(q) = 0$, the zero dynamics are enforced as follows:

$$\ddot{y} = L_f^2 y + L_g L_f y \cdot u = 0 \tag{2}$$

where L_f and L_g are the Lie derivatives that encode the system's nonlinear dynamics, and u is the control input. In a zero-dynamics system, when the input of the system is zero, the output remains zero or converges to values close to zero. This zero-dynamics behavior typically indicates system stability and convergence.

In the HZD controller, the goal of achieving an output function of zero is achieved through the design of virtual constraints. Virtual constraints are imposed via feedback control on the configuration variables of the robot, driving the output function to zero. The hybrid nature of walking requires additional conditions for impact events, expressed through the hybrid invariance condition:

$$y^-(q) \cdot y^+(q) \leq 0 \tag{3}$$

where the superscripts – and + denote the pre- and post-impact states, respectively. Stable gait control was achieved by selecting appropriate control variables and constraint designs.

Jessy Grizzle at the University of Michigan has primarily adopted HZD [49], which has been successfully applied to a series of robots, including MABEL [50], Cassie [51], and RABBIT [52]. Hereid *et al.* combined trajectory optimization methods with a multi-domain HZD control framework and then applied this approach to the underactuated spring-legged humanoid robot DURUS [53]. The experimental results demonstrate the ability to quickly and reliably generate efficient multi-contact robot walking gaits.

2.2.3. MPC

MPC is an online optimization control method used to achieve optimal control of dynamic systems. The principle of MPC can be summarized as follows: first, a mathematical model of the system needs to be established. In the context of humanoid robots, this typically involves building a control-oriented dynamic model based on the established kinematic and dynamic models of robots. Next, the system model is used for state prediction, iteratively predicting the system's future states and output variables based on future input variables. An optimization problem is then solved to select the optimal control strategy, considering constraints such as input limits and state-variable constraints.

The MPC optimization problem is formulated as follows:

$$\begin{aligned} &\min_{u_{0:N-1}} \sum_{k=0}^{N-1} \left(\|x_k - x_{\text{ref}}\|_Q^2 + \|u_k\|_R^2 \right) + \|x_N - x_{\text{ref}}\|_P^2 \\ &\text{s.t.} \quad x_{k+1} = f(x_k, u_k), \quad x \in \mathcal{X}, \quad u \in \mathcal{U} \end{aligned} \tag{4}$$

where Q, R, and P are weighting matrices that prioritize different aspects of the control performance, N is the prediction horizon, and $\mathcal{X}$ and $\mathcal{U}$ represent state and input constraints, respectively.

MPC is an iterative process in which the controller periodically predicts, optimizes, and controls the system to continuously and dynamically adjust it. A classic example of applying MPC to bipedal robots was presented by Li and Quan [54], who proposed a new framework for dynamic legged robot MPC based on forces and torques. In their work, a 10-degree-of-freedom bipedal robot was simplified into a rigid-body dynamics model. Using

this model, they established a predictive model for the MPC control system. By minimizing state errors and controlling input variations, an optimization problem was formulated to determine the optimal ground reaction forces and torques. These forces and torques were then mapped to the torque of the five actuators on each leg to generate the desired inputs. The optimal inputs computed via MPC help the robot achieve precise foot placement during swing-leg motion, thus avoiding instability and oscillations. The proposed MPC framework enables precise velocity tracking for 3D dynamic movements and is applicable to dynamic motions such as fast walking, jumping, and running using the same control parameters. This method also exhibits robustness on rugged terrains.

2.2.4. Whole-body control

Whole-body control (WBC) can be understood as a method for dynamically coordinating and controlling multiple tasks, or more explicitly, as prioritized multi-task control. The concept of executing prioritized multi-task control first appeared in the 1980s, pioneered by Sentis and Khatib, who applied this approach to humanoid robots and termed it WBC[55,56]. The fundamental WBC optimization problem is formulated as follows:

$$\min_{\tau,f} \sum_i w_i \| J_i \ddot{q} + \dot{J}_i \dot{q} - a_i^{des} \|^2 \tag{5}$$

subject to the following dynamic constraints:

$$M(q)\ddot{q} + C(q,\dot{q}) + g(q) = S^T \tau + J_c^T f \tag{6}$$

and contact constraints

$$J_c \ddot{q} + \dot{J}_c \dot{q} = 0 \tag{7}$$

where $M(q)$ is the mass matrix, $C(q,\dot{q})$ represents Coriolis and centrifugal forces, $g(q)$ is the gravity vector, J_c is the Jacobian for task i, and f represents contact forces.

Luis Sentis and Oussama Khatib introduced WBC as a framework based on task and posture spaces to enable multitasking behaviors in humanoid robots. The task space refers to the specific objectives that the robot must accomplish, such as grasping or walking, whereas the posture space corresponds to the joint angles or positions that control the robot's posture. The relationship between these spaces is established through projection and mapping, where projection maps the robot dynamics into task-relevant spaces, and mapping transforms task-space commands into posture-space control inputs. The WBC aims to simultaneously control multiple tasks by prioritizing them according to their importance.

Moro and Luis further classified WBC controllers[57] based on the output type and problem-solving method. Output-wise, controllers are velocity-based (producing joint velocities) or torque-based (producing joint torques), with these outputs ultimately converted into motor currents via low-level impedance or torque-feedback controllers. Regarding solution methods, the controllers are either closed-form or optimization-based. Closed-form controllers derive outputs through algebraic operations (such as projection and matrix inversion), allowing a fast hierarchical resolution of task priorities without heavy computation. Optimization-based controllers solve an optimization problem to find solutions, which generally requires more computation time.

Kim *et al.* proposed a controller combining WBC with MPC and applied it to the Mini-Cheetah quadruped robot[58]. In this framework, the MPC computes the optimal distribution of reaction forces over an extended horizon using a simplified model, whereas the WBC generates joint torque, position, and velocity commands based on the MPC outputs. Unlike traditional WBC approaches, this controller emphasizes reaction force

Table 2. Comparison of dynamic control methods in humanoid robots

Method	Key characteristics	Cases	Multidimensional insights
HZD	• Virtual constraint enforcement • Efficient with compliant legs (e.g., springs)	• Cassie walking at 2.1 m/s on flat terrain • Soft/uneven surfaces (sand)	• Low computational burden (offline trajectory optimization) • High energy efficiency • Limited adaptability to unknown disturbances • Requires accurate modeling of underactuated dynamics
MPC	• Receding-horizon QP solving • Real-time disturbance handling	• Terrain-adaptive walking (slope, gravel, steps) with ALIP-based Cassie • Fast walking at 1.6 m/s	• Strong robustness to terrain variations • High computational demand • Sensitive to model errors • Supports constraint-based planning
WBC	• Prioritized task-space optimization • ID + contact handling	• Atlas parkour (jumping, flipping, handstands) • Balance + manipulation synergy	• Enables complex coordinated behaviors • Requires full-body dynamic/contact models • Sensitive to latency and sensor delays • Less suitable for high-speed gaits

HZD: Hybrid zero dynamics; MPC: model predictive control; QP: quadratic programming; ALIP: angular mo-mentum linear inverted pendulum; WBC: whole-body control; ID: inverse dynamics.

commands, thereby enabling high-speed dynamic maneuvers during aerial phases. It was validated with six different gaits in various environments, including outdoor and treadmill tests, and achieved a top speed of 3.7 m/s. Additionally, Zhu *et al.* proposed a dynamic cutoff frequency scheme to balance the speed and accuracy[59].

2.3. Sum

This section presents a dynamics-based control framework for humanoid robots, leveraging simplified models such as the Cart-Table, LIP, LIPP, and SLIP to reduce computational complexity while maintaining stability in bipedal locomotion. Control strategies include PID for basic balance and three widely used approaches: HZD, MPC, and WBC. Table 2 provides a comparison of these three commonly used methods.

- High-speed, multi-task scenarios: Combine MPC for agile trajectory planning with WBC for coordinated whole-body execution.
- Challenging terrains: Use HZD with compliant mechanisms to enhance adaptability and energy efficiency.
- Hardware-driven choices: Prefer HZD for passive compliance; use MPC or WBC when active torque control is needed.

By combining simplified models with complementary control strategies, this integrated approach enables stable, efficient, and versatile locomotion for diverse humanoid robot applications.

3. LEARNING-BASED CONTROL METHODS

Learning-based control methods for humanoid robots include RL[60], imitation learning, phase-based methods, and large language models (LLMs). RL optimizes robot actions in complex environments through trial-and-error and reward feedback, thereby enabling autonomous adaptation and improvement. Imitation learning accelerates strategy acquisition by leveraging expert demonstrations and enhancing training efficiency through direct mimicry of human behavior. Phase-based methods capture periodic and pseudo-periodic patterns in human motion, allowing robots to replicate natural rhythmic locomotion with high fidelity. LLMs trained on vast datasets support multi-task and transfer learning, boosting robots' perception and decision-making by providing a broad contextual understanding. Integrating these advanced learning approaches significantly enhances the autonomy, adaptability, and intelligence of humanoid robots, enabling them to perform complex tasks with improved efficiency and accuracy.

3.1. RL

RL[61–63], a key area within machine learning (ML), is fundamentally based on Markov decision processes (MDPs), defined by six main parameters $\langle S, A, T, R, C, \rho \rangle$. Here, S and A denote the state and action spaces,

Table 3. Legged robot tasks using RL

Categorization	Method	Ref.	Pros	Cons
Model-free methods	Q-Learning	[65], [66], [67]	• Simple, model-free • Model-free learning • Simple and straightforward • Convergence guarantees • Off-policy learning • Suitability for discrete action spaces	• High-dimensional inefficiency • Scalability issues • Exploration-exploitation dilemma • Learning rate sensitivity • Temporal-difference error accumulation • Inefficiency in continuous spaces
	SARSA	[68], [69], [70]	• On-policy learning • Policy-sensitive • Convergence • Suitable for dynamic policies • Handles stochastic environments	• Potentially slower learning • Exploration-exploitation trade-off • Scalability issues • Sensitivity to learning parameters • Suboptimal policies in certain cases
Model-based methods	Dyna-Q	[71], [72]	• Accelerated learning • Efficient use of data • Reduced exploration costs • Improved sample efficiency • Flexibility and adaptability • Better generalization • Suitability for complex environments	• Model accuracy-dependence • Computational overhead • Complexity in implementation • Scalability issues • Potential for overfitting • Exploration-exploitation balance
Policy gradient methods	REINFORCE	[73]	• Direct policy optimization • Continuous action spaces	• Sample inefficiency • High-variance in gradients
	Actor-critic	[74], [75], [76]	• Stochastic policies • Suitable for high-dimensional data action spaces • Improved exploration • End-to-end learning • Efficient policy updates	• Sensitive to hyperparameters • Computationally intensive • Risk of suboptimal policies • Implementation complexity • Bias in value estimation
Deep reinforcement methods	DQN	[77], [78]	• Handles high-Dim. state spaces • End-to-end learning • Experience replay • Target networks • Scalability • Improved performance	• Sample inefficiency • Computationally-intensive • Exploration challenges • Memory requirements • Stability issues • Sensitivity to hyperparameters • Delayed rewards • Overfitting risk, data-hungry
	DDPG	[79], [80]	• Stable for continuous actions • Handles continuous action spaces • Deterministic policy • Efficient exploration with noise • End-to-end learning • Experience replay • Actor-critic framework • Target networks	• Increased complexity • Sample inefficiency • Hyperparameter sensitivity • Exploration challenges • Computational demands • Stability issues • Risk of overfitting • Implementation complexity
	PPO	[81], [82], [83]	• Stable, scalable • Stability and reliability • Improved sample efficiency • Ease of implementation • Robust performance • Continuous and discrete action spaces • Clipped surrogate objectives • Reduced computational complexity	• Hyperparameter sensitivity • Sample inefficiency • Hyperparameter sensitivity • Implementation complexity • Exploration challenges • Scalability issues • Long training times

RL: Reinforcement learning; SARSA: state-action-reward-state-action; DQN: deep Q-network; DDPG: deep de-terministic policy gradient; PPO: proximal policy optimization.

respectively; R and C represent the reward and cost functions, respectively; T is the state transition function; and ρ is the discount factor. In humanoid robot research, RL is frequently combined with deep-learning techniques. Hybrid methods that integrate model-based and learning-based approaches have also been developed to exploit the advantages of both methods. According to [64], deep RL frameworks for bipedal robots can be divided into two types: end-to-end and hierarchical, the latter featuring a layered control structure. Further details on this classification are beyond the scope of this study. The classification of RL methods is summarized in Table 3.

3.1.1. Model-free methods

3.1.1.1. Q-Learning

Q-Learning [65–67] is a classic model-free RL algorithm that learns discrete action-value mappings without requiring a model of the environment. For humanoid robots, Q-learning is suitable for discrete tasks such as basic navigation or simple motor control in environments with relatively simple dynamics. The Q-values are

updated iteratively using

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{8}$$

where $Q(s,a)$ denotes the value of taking action a in state s, α is the learning rate, r is the received reward, γ is the discount factor, s' is the next state, and a' is the next action.

3.1.1.2. State-action-reward-state-action

The state-action-reward-state-action (SARSA)[68–70] is another model-free method that updates the action value function using the agent's current policy. In humanoid robots, SARSA is well-suited for on-policy learning tasks, such as real-time balancing or locomotion, where the current policy is continuously updated during execution. Its update rule is given by

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma Q(s',a') - Q(s,a) \right] \tag{9}$$

The variables have the same meaning as in Equation 8, with the key difference being that SARSA uses the actual action a' chosen by the current policy instead of the maximum Q-value.

3.1.2. Model-based methods

Dyna-Q[72,84] is a hybrid method that integrates model-free Q-learning with a learned model of the environment. This enables both real and simulated experiences to update the policy, thereby improving the sample efficiency. In humanoid robotics, Dyna-Q has been applied to tasks such as dynamic balance and obstacle avoidance, where real-world interactions are combined with simulated rollouts to accelerate policy learning.

3.1.3. Policy gradient methods

3.1.3.1. REINFORCE

REINFORCE[73] is a fundamental policy gradient algorithm used to directly optimize policies in continuous-action spaces. Humanoid robots are particularly suitable for tasks requiring smooth, high-dimensional motor control, such as walking, running, and executing complex movements. The algorithm updates the policy parameters via gradient ascent to maximize the expected return. The update rule is:

$$\theta \leftarrow \theta + \alpha \cdot \nabla_\theta \left[\log \pi_\theta(a_t \mid s_t) \cdot G_t \right] \tag{10}$$

where θ denotes the policy parameters, α is the learning rate, and $\pi_\theta(a_t \mid s_t)$ is the probability of selecting action a_t given state s_t. The return G_t is defined as a discounted cumulative reward:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k},$$

where γ is the discount factor, and r_{t+k} is the reward at time $t + k$.

3.1.3.2. Actor-critic

Actor-critic methods[74–76] combine the strengths of value-based and policy-based approaches, making them well-suited for humanoid robots operating in high-dimensional continuous action spaces. These methods consist of an *actor* that updates the policy, and a *critic* that estimates the value function to guide the actor's learning.

The actor's update rule is given by

$$\theta_{\text{Actor}} \leftarrow \theta_{\text{Actor}} + \alpha_{\text{Actor}} \cdot \nabla_{\theta_{\text{Actor}}} \left[\log \pi_{\theta_{\text{Actor}}}(a_t \mid s_t) \cdot \delta_t \right] \tag{11}$$

where θ_{Actor} are the actor's parameters, α_{Actor} is the learning rate, and δ_t is the temporal difference (TD) error. The TD error serves as a feedback signal for adjusting the policy and is computed using the critic network as follows:

$$\theta_{\text{Critic}} \leftarrow \theta_{\text{Critic}} + \alpha_{\text{Critic}} \cdot \nabla_{\theta_{\text{Critic}}} \left[(r_{t+1} + \gamma V(s_{t+1}) - V(s_t))^2 \right] \tag{12}$$

Here, θ_{Critic} are the parameters of the value function approximator, α_{Critic} is the critic's learning rate, and $V(s_t)$ estimates the expected return from state s_t. The term $r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$ represents the TD error, which measures the discrepancy between the predicted and actual returns.

Actor-critic architectures are widely adopted in humanoid robotics owing to their sample efficiency and stability, particularly in dynamic and uncertain environments where real-time adaptation is essential.

3.1.4. Deep RL methods

3.1.4.1. Deep Q-network

The deep Q-network (DQN)[77,78] extends Q-learning by using deep neural networks to approximate the Q-function, enabling RL in high-dimensional state spaces. In the context of humanoid robots, DQN is suitable for tasks involving complex navigation and visual perception, such as obstacle avoidance and high-level decision-making. The Q-network is trained to minimize the difference between the predicted and target Q-values using the following loss function:

$$L = \mathbb{E}\left[(y - Q(s, a; \theta))^2 \right] \tag{13}$$

where $y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ is the target Q-value, θ are the current network parameters, and θ^- represents the parameters of a target network updated periodically to stabilize training.

3.1.4.2. Deep deterministic policy gradient

Deep deterministic policy gradient (DDPG)[79,80] is an off-policy, actor-critic algorithm designed for continuous action spaces. It combines deterministic policy gradients with deep function approximators, making it particularly effective for humanoid robots performing complex motor tasks, such as manipulation, locomotion, and precise coordination. DDPG uses two networks: an actor $\mu(s; \theta_{\text{Actor}})$ to represent the policy and a critic $Q(s, a; \theta_{\text{Critic}})$ to estimate the action-value function. The critic is trained to minimize the following loss:

$$L_{\text{Critic}} = \mathbb{E}\left[\left(r + \gamma Q(s', \mu(s'; \theta_{\text{Actor}}); \theta^-_{\text{Critic}}) - Q(s, a; \theta_{\text{Critic}}) \right)^2 \right] \tag{14}$$

The actor network is updated by maximizing the expected return using the deterministic policy gradient:

$$\nabla_{\theta_{\text{Actor}}} J \approx \mathbb{E}\left[\nabla_a Q(s, a; \theta_{\text{Critic}}) \nabla_{\theta_{\text{Actor}}} \mu(s; \theta_{\text{Actor}}) \right] \tag{15}$$

Table 4. Legged robot tasks using IL

Method	Ref.	Pros	Cons
BC	[103], [104], [105]	• Simplicity and ease of implementation • Rapid learning • Efficiency in supervised settings • Reduced risk • High-quality performance	• Dependence on quality and quantity of data • Limited generalization • Overfitting • Lack of exploration • Error propagation • Noisy or suboptimal expert demonstrations
IRL	[101], [106], [107], [108]	• Understanding of expert behavior • Generalization • Robustness to suboptimal data • Learning from sparse data • Transferability • Interpretable reward discovery	• High complexity and computational cost • Ambiguous reward functions • Dependence on expert quality • Long training time • Overfitting risk • Sensitivity to demonstration variations
GAIL	[102], [109], [110] [111], [112], [113], [114]	• High-quality imitation • Robust to suboptimal demonstrations • No need for explicit rewards • Effective generalization • Efficient use of limited data • Learns from interactions	• Training instability • High computational complexity • Mode collapse • Sensitive to expert quality • Ambiguous learned behaviors • Requires careful tuning

IL: Imitation learning; BC: behavior cloning; IRL: inverse reinforcement learning; GAIL: generative adversarial imitation learning.

3.1.4.3. Proximal policy optimization

Proximal policy optimization (PPO)[81–83,85–89] is a widely used on-policy algorithm that stabilizes training by limiting policy updates through a clipped surrogate objective. It is well suited for humanoid robots performing tasks that demand robustness and adaptability, such as multijoint coordination, variable terrain walking, and dynamic interaction. The clipped objective function used by PPO is as follows:

$$L_{\text{Clip}} = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) A_t \right) \right] \tag{16}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the new and old policy, A_t is the advantage estimate, and ε is a hyperparameter that controls the update range to ensure conservative policy improvement.

3.1.4.4. Hierarchical reinforcement learning

Deep RL methods can be further categorized into end-to-end and hierarchical methods. Hierarchical reinforcement learning (HRL)[90–97] decomposes complex tasks into simpler subtasks, each governed by a local policy, with a higher-level policy coordinating the overall behavior. This framework improves learning efficiency and task generalization, particularly for humanoid robots engaged in sequential or compound tasks, such as object manipulation, long-horizon navigation, and whole-body coordination.

3.2. Imitation learning

Imitation learning[98,99] enables humanoid robots to learn and perform complex tasks rapidly by replicating human demonstrations. Leveraging high-quality demonstration data allows robots to accurately imitate actions, significantly reducing the training time while enhancing control robustness and effectiveness. This approach improves the adaptability of robots in dynamic or unfamiliar environments, making it a vital technology for advancing robot intelligence. Key imitation learning methods include behavior cloning (BC),[100] inverse RL,[101] and generative adversarial imitation learning (GAIL)[102]. The classification of imitation learning approaches for humanoid robots is presented in Table 4.

3.2.1. BC

BC[104,105] is a basic imitation learning approach formulated as a supervised learning problem. It models expert behavior by learning a mapping from the observed states to actions using expert trajectories. Given a

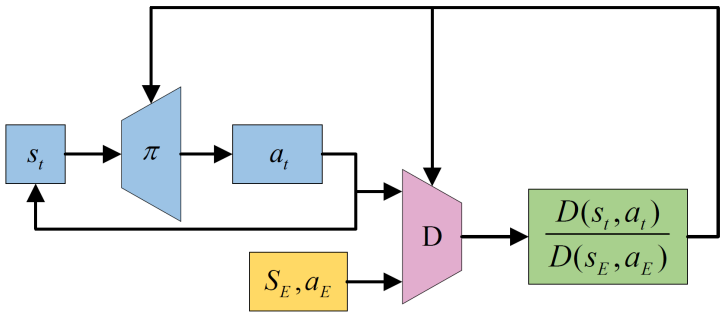


Figure 3. The general framework of GAIL. GAIL: Generative adversarial imitation learning.

discretized dataset of n expert demonstrations, the algorithm directly feeds state-action pairs into a neural network to train the policy. Although BC was initially applied to autonomous driving, it suffers from compounding errors owing to the lack of feedback, limiting its effectiveness in complex or long-horizon tasks. Consequently, BC has seen limited advancement in humanoid robot imitation learning.

3.2.2. Inverse reinforcement learning

Inverse reinforcement learning (IRL)[101] reverses the typical RL process by first inferring the reward function from expert demonstrations and then learning a policy using that reward. This addresses a key limitation in traditional RL: the difficulty in designing effective reward functions, which significantly affects the control performance. The input to the IRL is a set of expert-generated trajectory samples, denoted as $T_E = \{\tau_1, \tau_2, \dots\}$, where each τ_i is an ordered sequence of state-action pairs from expert demonstrations.

$$T_E = \{s_1, a_1, s_2, a_2, \cdots | \tau_i\} \tag{17}$$

IRL aims to derive a reward function from expert demonstration samples. It posits that trajectories generated by the expert policy yield maximal rewards, thereby necessitating the reward function r^* to satisfy the inequality $E_{\pi_E}[r^*(s,a)] \geq E_{\pi}[r^*(s,a)]$. By reformulating this inequality into an optimization problem, the process of determining the reward function unfolds as follows:

$$\min_r \max_\pi E_\pi[r(s,a)] - E_{\pi_E}[r(s,a)] \tag{18}$$

Owing to the high-dimensional nature of humanoid robot control, computing reasonable reward functions using IRL requires significant computational power. Moreover, there is a possibility of deriving multiple reward functions that yield similar outcomes, necessitating further expert analysis. These challenges significantly limit the application of IRL in humanoid-robot control.

3.2.3. GAIL

Figure 3 illustrates the principle of GAIL[102]. GAIL integrates imitation learning with generative adversarial networks (GANs)[109], reformulating learning from expert demonstrations within the GAN framework. This approach addresses the compounding errors in BC caused by limited environmental interaction. In GAIL, the policy, which maps states to actions, acts as the generator, whereas the reward function, which evaluates state-action pairs (s, a) and outputs instantaneous rewards, serves as the discriminator. Policy optimization is performed based on the current reward function, which is analogous to generator training, whereas optimizing the reward function corresponds to training the discriminator. By adopting this GAN architecture, GAIL enhances robustness, representational power, and computational efficiency compared to traditional imitation learning methods[102].

In addition, GAIL models the relationship between the generator and discriminator as an adversarial game process, thereby avoiding the explicit process of solving for the reward function. The objective function of this game is expressed as follows:

$$\min_{\pi} \max_{D} E_{\pi}[\log D(s,a)] + E_{\pi_E}[1 - \log D(s,a)] - \lambda H(\pi) \tag{19}$$

where the discriminator $D(s,a)$ aims to distinguish between decision trajectories (s,a) generated by the expert policy E_{π_E} and those produced by the policy being optimized, similar to the role of the discriminator in GANs. $H(\pi)$ represents the entropy of the policy, maximizing the diversity of policy outputs to encourage exploratory behavior.

GAIL is the most widely used imitation learning method for humanoid-robot control. Peng[110] used motion capture data from public Mocap datasets (CMU and SFU) for low-level control by mapping motions to robot joints, while high-level control set target tasks, enabling robots to perform predefined actions and mimic human motions. Combined with unsupervised RL, GAIL can train skill libraries from large datasets for diverse tasks[111]. The adversarial motion priors (AMP) algorithm was validated on the Adam robot, tested in the Isaac Gym and Webots simulators, and deployed on hardware with highly human-like walking[114]. Reference[105] improved the rewards by combining task and imitation objectives using a multi-expert network for smoother behavior. Dai[112] proposed a benchmark with expert datasets and multiple-robot platforms. Unlike most prior studies using full-body motions,[115] mapped only upper-body motions onto the Unitree H1 robot, freeing the legs for diverse actions. Despite its successes, GAIL training can be unstable;[116] addressed this by integrating diffusion models to enhance reward smoothness and robustness. GAIL has also been effectively applied to quadruped robots[97,113,117].

3.3. Representation learning

Reference trajectories (e.g., motion capture data) have greatly advanced motion learning[110,118,119], but these methods struggle to generalize beyond the training-data distribution[117]. This limitation stems from imitation learning, which focuses on replicating trajectories rather than understanding the underlying dynamics, often resulting in memorization rather than true learning. Large distribution gaps between different motions hinder smooth transitions, and directly linking motion to the original trajectories leads to inefficient representations[120]. A more systematic approach to leverage the spatial structure of motion is required.

Structured representations introduce inductive biases to manage complex motions efficiently by extracting fundamental features and temporal dependencies[121,122]. These representations enable the generation of continuous and diverse motions[123]. Classical controllers often parameterize motion using cyclic sine-wave trajectories[124] or CPGs. Self-supervised models, such as autoencoders, learn latent motion dynamics, enabling complex decision-making[125], long-term tasks[126], and motion imitation[127].

Frequency domain methods, as shown in Figure 4, are another structured approach widely used in motion tasks. The periodic Autoencoder (PAE) applies frequency transformations as inductive biases to improve the smoothness and naturalness of motion transitions[128]. The latent parameters of PAE serve as effective holistic states for downstream tasks[123] but are limited to local frames. Fourier latent dynamics (FLD), an extension of PAE, models spatiotemporal relations in periodic motions via a predictive latent structure [Figure 4][129]. FLD efficiently represents high-dimensional trajectories in a continuous latent space, capturing fundamental features and temporal dependencies, and excels in robust long-term motion learning while avoiding unlearnable regions.

3.4. LLMs

LLM technology is a deep learning–based natural language processing approach that trains large-scale models capable of understanding and generating natural language text. These models can produce codes or support

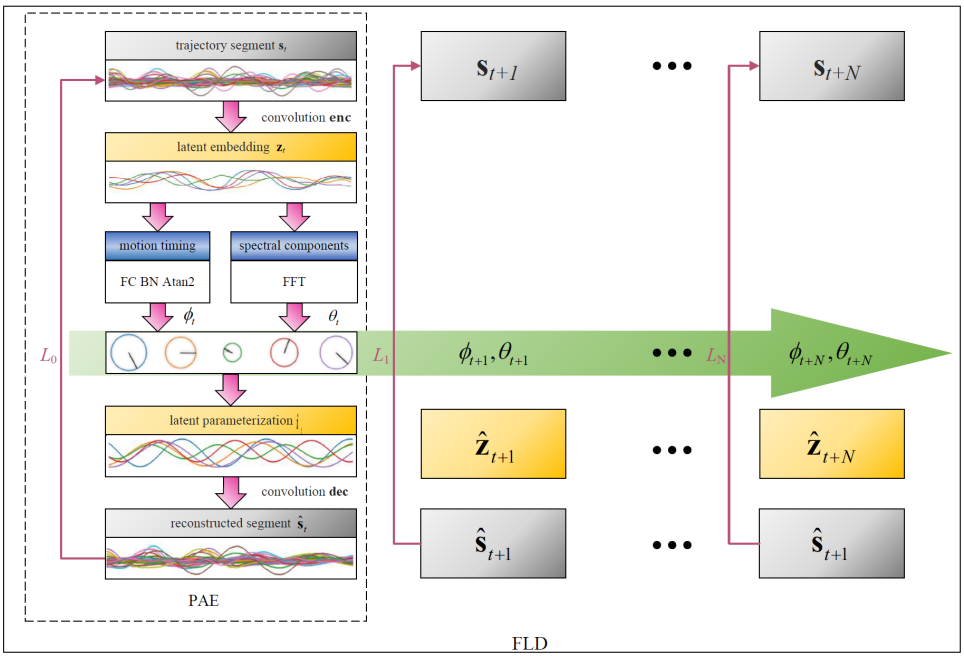


Figure 4. Structure diagram of PAE and FLD. PAE: Periodic Autoencoder; FLD: Fourier latent dynamics.

Table 5. Legged robot tasks using LLM

Application scenarios	Tasks	Works
Robotics	Language-conditioned imitation learning	[139], [140]
	Language-assisted RL	[141], [142], [143], [144], [145], [146]
	Language-image goal-conditioned value learning	[147], [148]
	High-level task planning	[149], [150], [151]
	Code generation based on LLM	[152], [153], [154], [155], [156], [157], [158], [159]
Human-computer interaction	speech interaction	[160], [161], [162], [163], [164], [148], [165]

LLM: Large language model; RL: reinforcement learning.

common-sense reasoning, whereas visual language models enable open-vocabulary visual recognition[130]. The key advantage of LLMs lies in their ability to learn rich linguistic knowledge and patterns from massive text corpora, allowing them to grasp semantics, grammar, and other language aspects with enhanced processing capabilities.

LLMs have achieved significant breakthroughs in both language and vision domains, exemplified by models such as BERT[131], GPT-3[132], Codex[133], GPT-4[134], SAM[135], DALL-E 2[136], and CLIP[137]. In robotics, LLM applications can be broadly divided into direct robotic integration and related applications[138]. This section focuses on research involving LLMs in legged robots.

The main applications of LLMs in legged robot tasks are summarized in Table 5, with robot policy learning and code generation being particularly important. In the previously mentioned GAIL research, adversarial imitation enabled robots to progressively align their motion trajectories with reference trajectories, although it often required task-specific strategies. To overcome this, combining LLMs with GAIL allows the pre-training of diverse motions via GAIL, while LLMs act as strategic planners to execute specific action sequences[139,166]. Moreover, LLMs play vital roles in semantic analysis and code generation. By leveraging their extensive knowledge, control tasks can be fully delegated to LLMs, which can be modularized into multiple specialized roles, such as task analysis, code generation, and debugging[153].

Table 6. Representative cases of humanoid robot learning applications

Case	Method type	Key metrics/phenomena	Multidimensional insights
DeepMind Soccer Robot (Deep RL)	End-to-end RL	Compared to scripted baseline: walking speed 181%, turning speed 302%, get-up time 63%, kicking speed 134%	• High autonomy & environmental adaptability • Large sample consumption (tens of millions of steps) • Zero-shot Sim-to-Real transfer proves the effectiveness of domain randomization
Tesla Optimus – Household Tasks	RL + Behavior Library	Already able to perform sequential actions such as "cut vegetables – plate – wipe table"	• Uses high-frequency visual closed-loop control; high computational load • Factory simulation helps shorten deployment time; better sample efficiency than pure RL • Safety mechanism: self-supervised detection of dangerous poses with co-control stop
FLD Representation Learning (MIT Humanoid)	Representation Learning	Latent space interpolation error < 0.04 rad; transitions smooth even for unseen trajectories	• Compression ratio ≈ 70%; inference latency < 1 ms • Significantly improves multi-action switching smoothness; enables long-horizon tasks • No need for reward engineering; minimal demonstration required
LUMOS – Language-Conditioned Imitation	LLM + IL	CALVIN multi-task success rate improved from 47% 63%; zero-shot transfer to real robot arm	• Strong generalization; capable of understanding complex language • High sample efficiency: < 1% labeled language data • World model training narrows the sim-to-real gap, but inference requires GPU

RL: Reinforcement learning; FLD: Fourier latent dynamics; LLM: large language model; IL: imitation learning; GPU: graphics processing unit.

3.5. Applications of learning-based methods in humanoid robots

Learning-based methods are widely used in humanoid robot motion control, with RL being the most common approach that processes proprioceptive observations (e.g., joint angles and velocities) and outputs joint actions through proportional-derivative (PD) control for locomotion. Supervised learning with encoders can further enable terrain adaptation by extracting the environmental features. For more human-like motions, imitation learning methods, such as GAIL, align robot movements with motion-capture data through adversarial training, as demonstrated by Tesla Optimus learning fine manipulation from human demonstrations. Representation learning (e.g., PAE, FLD) encodes motion primitives in latent space for efficient movement generation, whereas LLMs serve as high-level planners that interpret natural language commands to coordinate low-level controllers, effectively acting as the robot's "brain." As demonstrated in Table 6, we compiled real-world application examples of these methods.

3.6. Sum

The field of humanoid robot control has undergone a transformative shift from traditional rule-based approaches to data-driven learning paradigms. By integrating various ML techniques, modern control methods have significantly enhanced the environmental adaptability and task execution capabilities of robots. RL enables autonomous policy optimization in complex environments using reward mechanisms. Imitation learning rapidly acquires operational skills by mimicking the demonstrations of experts. Deep neural networks achieve high-precision control through big data training, whereas representation learning automatically extracts key features from raw data to support intelligent decision making. These cutting-edge technologies collectively advance humanoid robots toward higher levels of flexibility, autonomy, and intelligent performance.

Meanwhile, we compared four learning-based methods–RL, imitation learning, representation learning, and LLMs–across six dimensions: autonomy, sample efficiency, environmental adaptability, computational cost, interpretability, and safety, with the results shown in Figure 5. RL demonstrates outstanding performance in autonomous exploration and environmental adaptation but requires extensive trial and error and has lower safety. Imitation learning relies on high-quality demonstration data while maintaining a balanced overall performance. Representation learning exhibits high sample efficiency and good interpretability, making it

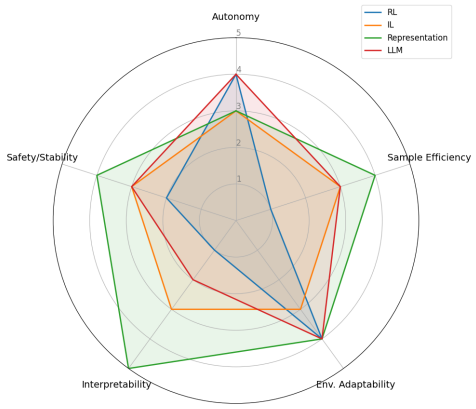


Figure 5. Comparison of learning methods.

suitable for structured tasks. LLMs show strong generalization and adaptability but suffer from high computational costs and limited interpretability. Therefore, these methods can be strategically combined across certain dimensions to leverage their respective strengths and compensate for their weaknesses.

As shown in Table 7, although each control method has unique advantages, learning-based approaches demonstrate particularly promising potential. Although traditional methods (e.g., MPC and WBC) have certain advantages in motion control, learning-based methods are increasingly becoming the key solution for advanced challenges, such as navigating complex terrains, autonomous fall recovery, and dynamic object manipulation, as application scenarios continue to expand. Notably, learning-based methods also offer lower technical barriers and shorter development cycles, which not only reduce industry entry thresholds but also accelerate innovation across the humanoid robotics field. With continuous advancements in algorithms and computing hardware, learning-based control methods are poised to lead the next generation of technological revolutions in humanoid robotics.

4. PATH PLANNING AND OBSTACLE AVOIDANCE

An essential capability of mobile robots is autonomous path planning and obstacle avoidance in complex dynamic environments[177–179]. Path planning seeks an optimal route from a start to a target point, optimizing criteria such as the shortest distance, minimum travel time, or lowest operational cost while successfully avoiding obstacles. As effective path planning inherently requires obstacle avoidance, these two challenges are closely linked.

Environmental perception and autonomous localization are critical for motion planning. Robots employ sensors, such as laser scanners and cameras, to collect environmental data, build local maps, and simultaneously estimate their position within these maps using simultaneous localization and mapping (SLAM) technology[180]. SLAM enables rapid and autonomous localization and mapping in unknown environments, thereby supporting precise path planning and obstacle avoidance. SLAM methods are classified based on the sensor type, as shown in Figure 6.

- 2D Laser SLAM:
  Rotating laser scanners are used to measure the distance and direction in a plane. They enable localization and mapping but only capture 2D planar information, limiting their use to applications such as floor-cleaning robots or drones operating in a single plane and not suited for taller robots such as humanoids.
- 3D Laser SLAM:
  It provides detailed 3D perception, including height and shape, but requires expensive equipment and

Table 7. Control methods and their advantages/disadvantages for humanoid robots

Control method (category)	Basic principle	Refs.	Advantages	Disadvantages
IDC	Computes desired torques based on ID for tracking target trajectories	[167]	Simple implementation, high trajectory accuracy	Sensitive to model errors
Contact-constrained ID control	Models ground or arm contacts as force or position constraints	[168]	Supports multi-contact, improves steady-state control	Complex contact modeling
OSC	Defines targets in task space (e.g., CoM or end-effector), then maps to joint space	[169]	Supports task prioritization, yields natural motion	Relies heavily on accurate Jacobian and mass matrix
Optimization-based control (QP/MPC)	Formulates control as an online constrained quadratic programming problem	[170]	Unified handling of multi-task and multi-constraint control	High computational demands for real-time execution
Forward dynamics + feedback control	Treats the system as a known model and designs stabilizing feedback law	[171]	Strong stability guarantees, clear control design	Strong dependence on accurate system modeling
ZMP control	Maintains the projection of CoM within the support polygon for stability	[172]	Suitable for cyclic walking, mature and well-studied method	Not suitable for unstructured or highly dynamic tasks
MPC	Predicts future states and optimizes control inputs accordingly	[173]	Predictive capability, strong stability	High computational cost
State feedback control + estimation	Estimates system states via sensors for closed-loop feedback control	[174]	High robustness, suitable for real-world applications	Critically depends on state estimation accuracy
RL + imitation learning	Learns policy through trial-and-error or human demonstrations	[175]	Highly adaptive to unstructured environments	Safety concerns, high training cost
Hybrid control architecture (Hierarchical)	High-level policy planning combined with low-level optimization or feedback control	[176]	Flexible and modular structure, robust performance	Complex system integration and synchronization

IDC: Inverse dynamics control; ID: inverse dynamics; OSC: operational space control; CoM: center of mass; QP: quadratic programming; MPC: model predictive control; ZMP: zero moment point; RL: reinforcement learning.

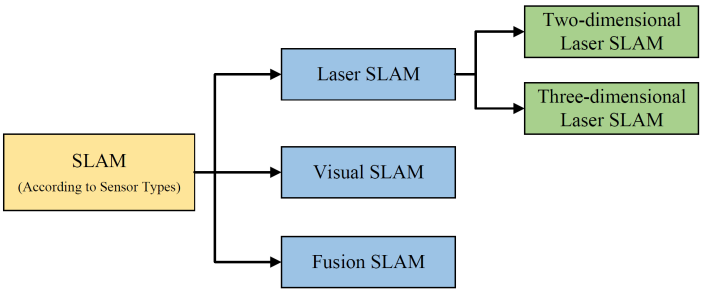


Figure 6. Classification of SLAM technology. SLAM: Simultaneous localization and mapping.

significant computational resources[181]. It is commonly used in indoor navigation, 3D modeling, and environmental monitoring.

- Visual SLAM:
Cameras are employed to extract image features and motion data for localization and mapping[182]. It is cost-effective compared to lidar, but is sensitive to lighting changes and texture scarcity, which may cause localization errors.
- Fusion SLAM:
It combines multiple sensors [e.g., lidar, cameras, and inertial measurement units (IMUs)] to improve accuracy and robustness. For example, ETH's Anymal quadruped robot integrates IMUs, lidar, and cameras[96], enhancing self-state perception and external environment mapping, respectively.

SLAM systems mainly consist of five components: sensor data acquisition, front-end odometry, loop closure

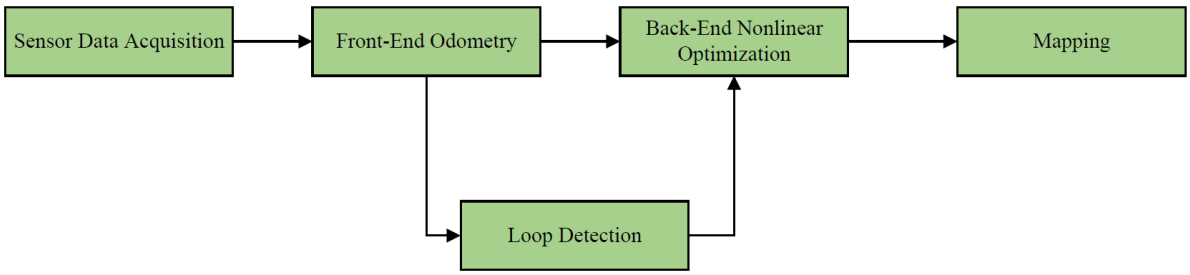


Figure 7. SLAM system framework. SLAM: Simultaneous localization and mapping.

Table 8. Common path planning algorithms

Types	Algorithm	Description	Pros	Cons
Graph search	Dijkstra [184]	Stepwise distance updates to find shortest paths	Guarantees shortest paths to all nodes	High computation; cannot handle negative edges
	A* [185]	Heuristic search combined with Dijkstra	Efficient; finds optimal paths	Performance depends on heuristic design
Sampling-based	RRT [186]	Random sampling to build exploration trees	Fast convergence; adapts to dynamic environments	Needs many samples
	PRM [187]	Builds roadmap in map space	Suitable for complex, high-dimensional spaces; adaptive	Path quality affected by sampling density
Optimization-based	D* [188]	Reverse search updating cost maps for path optimization	Efficient in dynamic environments	High memory usage for cost maps
	MPC [189]	Predictive model optimizing path with dynamic constraints	Handles multiple goals and constraints simultaneously	High computation; needs accurate models
Local obstacle avoidance	Artificial potential field [190]	Uses attraction and repulsion forces	Simple and real-time	Prone to local minima; parameter sensitive
	Dynamic window approach [191]	Samples surroundings to determine next motion	Real-time; adaptable; easy tuning	May get stuck in local minima; unsuitable for large areas

RRT: Rapidly-exploring random trees; PRM: probabilistic roadmap; MPC: model predictive control.

detection, back-end nonlinear optimization, and map construction [183], as shown in Figure 7.

The overall system framework of SLAM is shown in Figure 7. The SLAM process typically consists of the following steps: acquiring sensor data [e.g., images, light detection and ranging (LiDAR) point clouds, IMU data, and wheel encoder measurements]; front-end odometry estimates poses using tightly coupled visual, LiDAR, and inertial data; loop closure detection relocates the current frame to reduce cumulative drift; the back-end performs nonlinear optimization to minimize global errors and ensure consistency of the estimated trajectory and map; finally, the mapping module integrates sensor observations into a coherent global map.

Extensive research has been conducted on path planning and obstacle avoidance techniques for mobile robots, resulting in various mature algorithms. These algorithms can be broadly classified into four categories: graph-based, sampling-based, optimization-based methods, and local obstacle avoidance strategies. Table 8 provides a summary of the representative algorithms in each category along with their respective advantages and limitations.

Currently, research on humanoid robots primarily focuses on enhancing motion control and walking performance, with relatively little emphasis on trajectory planning. Humanoid robots can adopt trajectory-planning technologies from wheeled robots and autonomous vehicles [192]. Unlike wheeled robots, which must navigate around obstacles, humanoid robots can step over or traverse obstacles, offering novel insights into trajectory planning.

Boston Dynamics' ATLAS robot [193] uses LIDAR to scan the terrain, detect obstacles, and solve optimization problems for safe stepping region calculation and foot placement planning, which are fed into the controller.

The Iterative Regional Inflation Semidefinite programming (IRIS) algorithm computes obstacle-free convex polygonal regions, enabling both trajectory planning and obstacle avoidance. The parkour perception algorithm converts the sensor data into actionable information, whereas a model predictive controller computes the optimal movements. Additionally, an off-template movement controller simplifies the behavior generation by invoking actions from an action library[194].

Carnegie Mellon University's CHIMP humanoid robot constructs 3D environment models using multiview fusion and visual range systems, enhancing situational awareness with real-time texture mapping. Its geometric constraints are defined in Cartesian coordinates, and motion paths are generated using a constrained bidirectional rapidly exploring random tree algorithm[195]. CHIMP's planning architecture includes multiple planning nodes with a request/response interface to ensure a high-speed system response.

Honda's ASIMO robot[196] employs the A* search algorithm for optimal footstep sequencing and uses infrared sensors and CCD cameras for obstacle-avoidance. Delfin *et al.* proposed a vision-based navigation scheme integrating localization, planning, path tracking, and obstacle avoidance, focusing on 2D data to reduce computation[197]. Sahu *et al.* developed an adaptive particle swarm optimization algorithm to optimize humanoid robot trajectories by determining the best next step position[198]. Kusuma *et al.* applied the A* algorithm to a household assistant robot for indoor navigation, utilizing ultrasonic sensors for localization[199]. Kuffner *et al.* designed a motion planning algorithm based on rapidly exploring random trees, emphasizing bipedal humanoids' unique ability to step over obstacles[200].

These studies collectively demonstrate diverse approaches to navigation and obstacle avoidance in humanoid robots.

5. CHALLENGES AND FUTURE DIRECTIONS

5.1. Key challenges

Humanoid robotics faces three primary barriers to real-world deployment.

Cost Barriers: The sophisticated mechanical designs and sensor systems required[201] result in prohibitively high production costs at current manufacturing scales.

Hardware Constraints: While platforms such as Atlas demonstrate basic mobility, their limited degrees of freedom and 1-2 hour operational endurance restrict complex task execution[202].

Computational Demands: Implementing advanced controllers (e.g., Transformer-based approaches) requires substantial GPU resources[175], while maintaining real-time performance remains challenging.

5.2. Future directions

The potential of humanoid robots has not been fully explored. The following vital technologies are poised to enhance their capabilities in the future:

(1) Unified Frameworks: Presently, control frameworks for humanoid robots span multiple platforms, including Isaac Gym[203], Isaac Sim[204], Unity[205], MuJoCo[206], MuJoCo Playground[207], ROBOVERSE[208], each with distinct advantages. Isaac Gym leverages GPU capabilities for superior simulation performance, Unity offers a more user-friendly GUI for adjustments, and MuJoCo excels in simulation accuracy. Integrating these platforms to harness their collective strengths could streamline the user experience and maximize their utility.

(2) Environment-Based Learning: Most current humanoid robots lack advanced environmental perception and rely predominantly on adaptive learning methods, such as knowledge distillation, for navigation across various terrains. Incorporating LIDAR or vision-based systems to enhance environmental awareness could significantly improve the adaptability of humanoid robots in human-centric spaces.

(3) Integration with LLMs: One challenge in RL is the design of reward functions. Utilizing LLMs to generate a repository of reward functions and iterating through sampling can identify the optimal reward function, surpassing those designed by the experts. Additionally, the superior code generation capabilities of LLMs can boost the efficiency of robot development.

(4) Sim2Real Transition: Despite promising results in simulations, transitioning from simulations to real-world applications remains challenging. Although some frameworks demonstrate effective sim-to-real transfer, many studies are confined to simulation environments, underscoring the need for more robust and reliable methods for real-world implementation.

(5) Control algorithm–system synergy: The fusion of control algorithms should be tightly coupled with the specific characteristics of the controlled system. A promising research direction would be to leverage a profound understanding of bipedal robot dynamics to enhance the current control performance. This approach could bridge the gap between theoretical control models and practical implementation.

(6) Manipulation Performance: In real world, humanoid robots still struggle with various legged locomotion tasks, making advanced manipulation tasks involving arms and dexterous hands even more distant.

6. CONCLUSIONS

Humanoid robots have made significant progress in motion control and planning, with two dominant approaches emerging: dynamic-based and learning-based methods. Dynamics-based techniques, such as MPC and ZMP, provide stable and predictable control through precise modeling, whereas learning-based methods, such as RL and LLM integration, enable adaptive and intelligent behaviors. However, key challenges remain in bridging the simulation-to-real gap and reducing development costs. Future advancements are likely to focus on integrating these approaches with enhanced perception systems and dexterous manipulation capabilities. As these technologies mature, humanoid robots are poised to transition from research laboratories to real-world applications in domestic and industrial settings; however, significant engineering hurdles must still be overcome to achieve reliable and cost-effective deployment. The field's progress suggests that we may be approaching an inflection point where humanoid robots become practical tools rather than just research prototypes.

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Authors' contributions

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Conflicts of interest

All authors declared that there are no conflicts of interest.

Ethical approval and consent to participate

Not applicable.

Consent for publication

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